

Year 7 Tests (2023-24)

Ciaran Barnes

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Topic Test 1: Powers, Roots, BIDMAS and Negatives

1. Evaluate the following:

(a) 14^2

(b) 7^3

(c) $\sqrt{225}$

(d) $\sqrt[3]{729}$

2. Evaluate $(-5)^4$

3. Simplify $\sqrt{54}$

4. Evaluate the following:

(a) $-19 + (-23)$

(b) $-17 - (-27)$

(c) $17 \times (-5)$

(d) $-147 \div (-7)$

5. Evaluate $-375 \div 5^3 \times (17 - 4)$

Extension

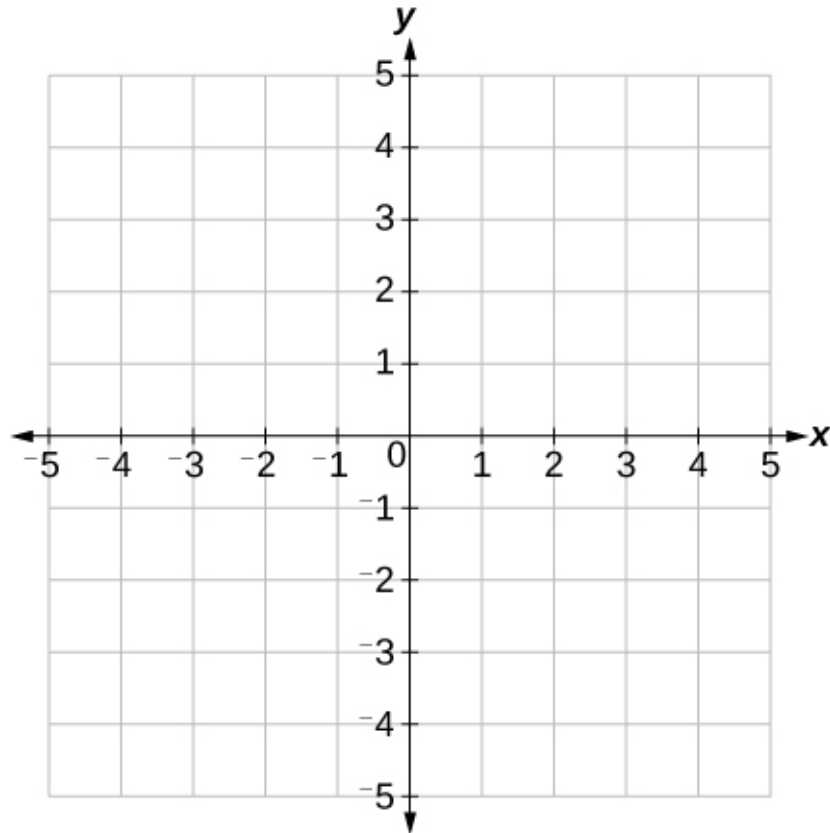
Evaluate the following:

$$\frac{\frac{65-11}{27} \times 3^3 - 3^2 \div 2 \times \sqrt{16}}{\left(2 \times \frac{18}{\sqrt[3]{216}}\right)^2}$$

Topic Test 2: Graphs, Coordinates and Primes

1. (a) On the following axes, mark the coordinates $(-2, 1)$, $(1, 0)$, $(0, -3)$.

(b) State a coordinate that would make the shape a square.



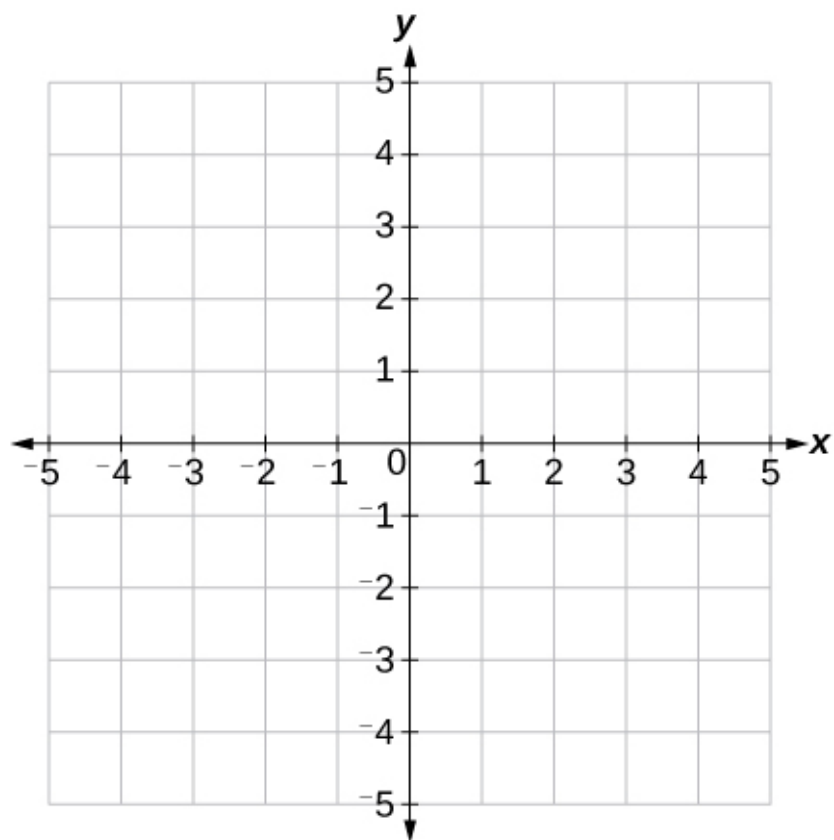
2. On the following axes, plot and clearly label the lines:

(a) $y = x$

(b) $y = -1$

(c) $y = -x$

(d) $x = 3$



3. Determine the prime factorisation of 4116.

4. The number 4356 is a square number. Prove that this is true and hence find its square root.

5. A number which is neither prime nor equal to 1 is called what?

Extension

The first and third digit, x , of the 5-digit number

$$x6x41$$

are the same. The number is divisible by 9. What is the number?

Topic Test 3: Fractions and Transformations

1. Evaluate the following leaving your answers as either improper or proper fractions:

(a) $\frac{3}{7} + \frac{7}{9}$

(b) $\frac{23}{24} - \frac{3}{8}$

(c) $\frac{3}{10} \times \frac{5}{8}$

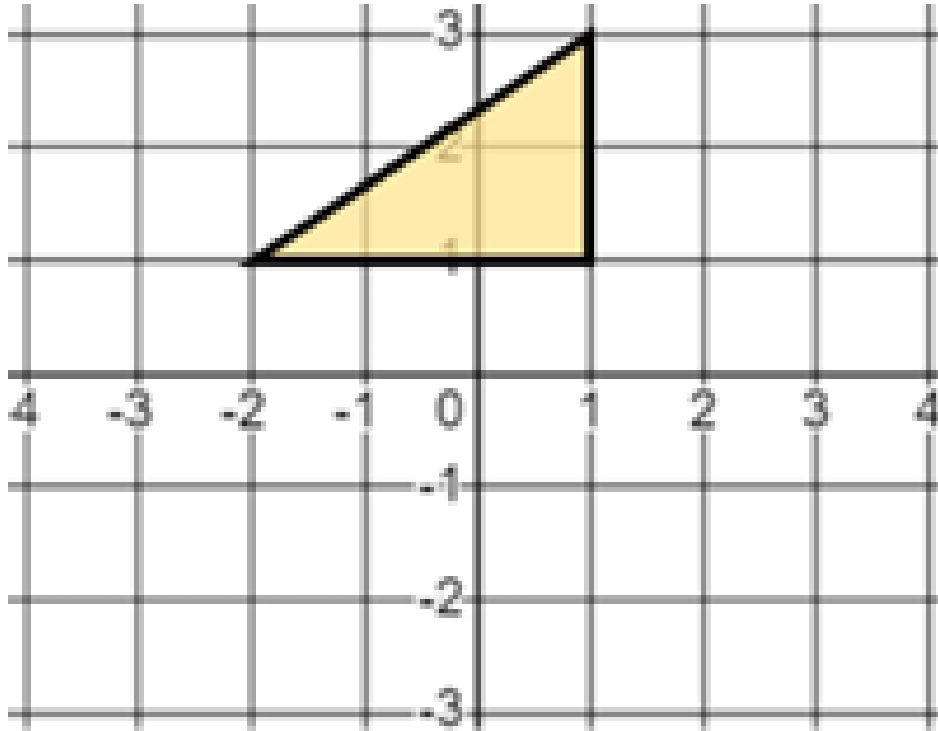
(d) $\frac{2}{7} \div \frac{4}{5}$

2. Perform the following transformations on the yellow triangle:

(a) Rotation by 180° about the point $(1,0)$. Label it A .

(b) Translation by the vector $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$. Label it B .

(c) Reflection in the y -axis. Label it C .



3. Evaluate the following leaving your answers as mixed numbers:

(a) $2\frac{1}{2} + \frac{3}{4}$

(b) $1\frac{5}{8} - 3\frac{4}{7}$

(c) $2\frac{1}{3} \times 3\frac{3}{4}$

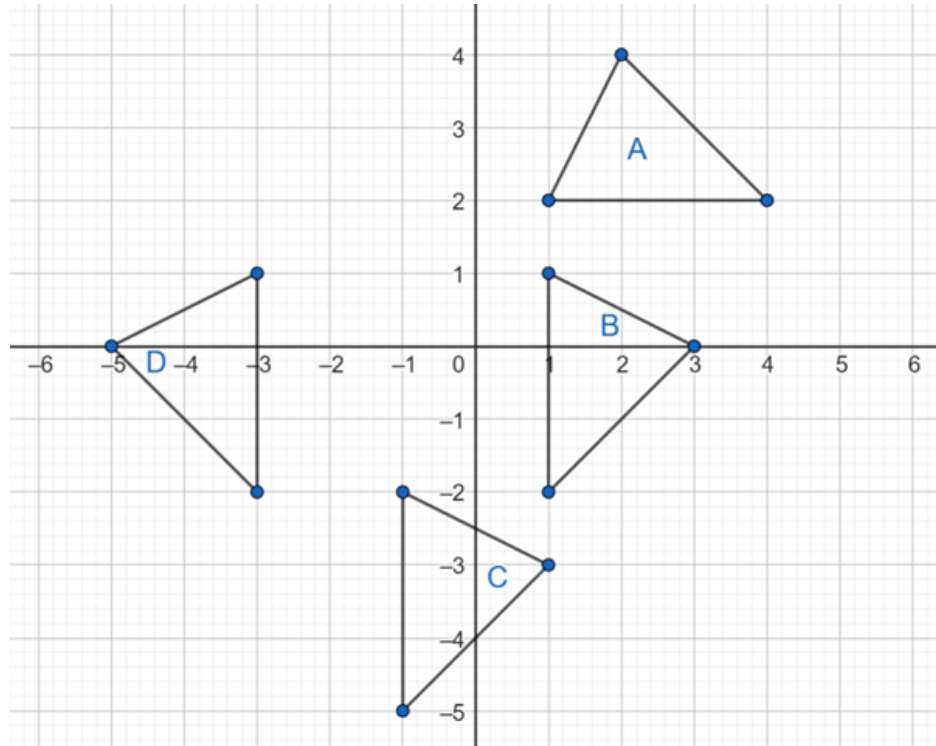
(d) $2\frac{5}{8} \div 1\frac{1}{6}$

4. Describe the following transformations as shown in the diagram:

(a) A to C

(b) D to B

(c) C to B



Extension

1. Evaluate

$$2 + \frac{1}{1 + \frac{1}{4 + \frac{1}{3}}}$$

2. Evaluate

$$2 \div (4 \div (6 \div (8 \div 10)))$$

Topic Test 4: Algebra and Geometry

1. Simplify the following expressions:

(a) $4(2a + 1)$

(b) $b(b - 3)$

(c) $2c(5 - 2c)$

(d) $-2d(d - 5)$

(e) $3x^2(2 - 6x)$

(f) $x(3y - 1)$

(g) $-2z^2(x - y)$

(h) $6x(x - 3x^2 + 1)$

2. A rectangle has one of its lengths 6cm longer than the other. The shorter length is denoted by x .

(a) Find and simplify an expression for the area of the rectangle.

(b) Find and simplify an expression for the perimeter of the rectangle.

3. Simplify the following expression:

$$3(1 + 4x) - 5(2 + 6x)$$

4. Consider the term $-3x^7$

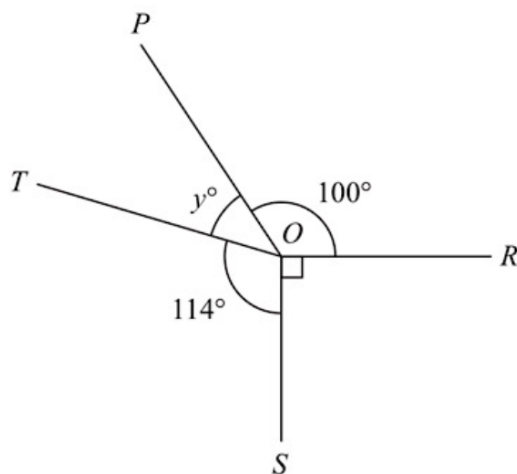
(a) What is the coefficient of this term?

(b) What is the index of this term?

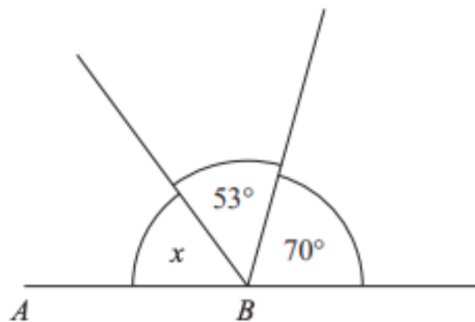
(c) What is the variable of this term?

5. Find the unknown angle in each case below. You do **NOT** need to give reasons for your answers but should show your working.

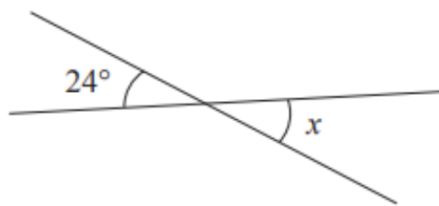
(a)



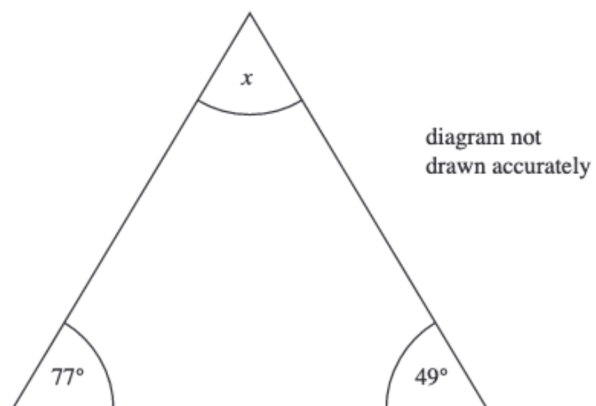
(b)



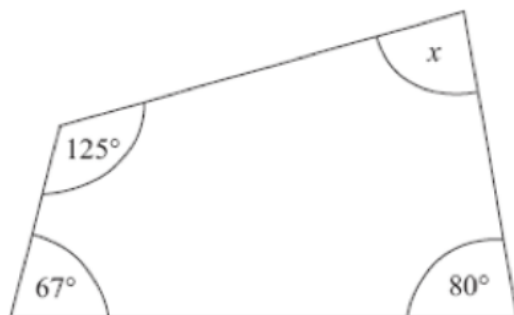
(c)



(d)



(e)



6. In each of the questions below, find the value of A by substituting the values for x and y into the given equation.

(a) $A = x - 5$ when $x = 3$

(b) $A = 3x + 1$ when $x = 2$

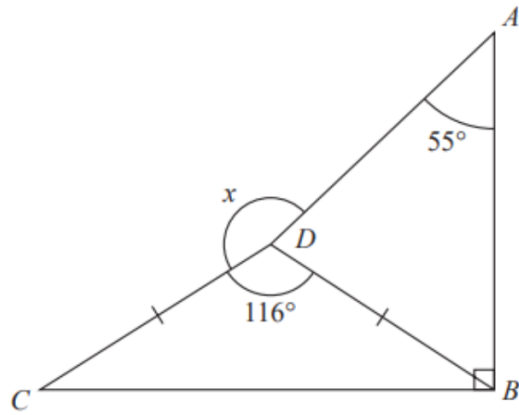
(c) $A = 2x^2 - 1$ when $x = -3$

(d) $A = 3x - 2y$ when $x = -7, y = 5$

7. In this question, you **MUST** give reasons for each stage of your working.

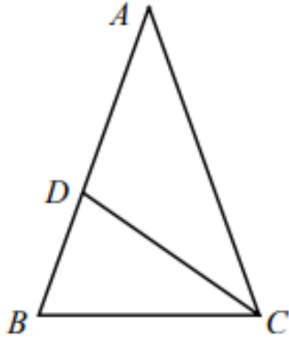
(a) Find the angle DCB .

(b) Find the angle x .



Extension

1. In the diagram below, you are given that the lengths $BC = CD$ and $AB = AC$. The line CD cuts the angle ACB in half. What is the size of angle CDA ?



2. Find the value of u when $x = 2$ if u is given by the equation

$$u = \frac{-3^x}{4} \div \frac{7}{2x-1}$$

Topic Test 5: Decimal Arithmetic and Solving Equations

1. Evaluate the following:

(a) $32.1247 + 3.21247$

(b) $0.0741 - 1.32$

(c) 13.1×3.69

(d) $0.144 \div 2.4$

2. Solve the following equations. Leave your answers as improper or proper fractions as necessary.

(a) $2a + 4 = 10$

(b) $2(b + 3) = 6$

(c) $8 - 7c = -6$

(d) $3(2 - d) = 24$

(e) $5w - 5 = 2 = 3w$

(f) $4(2x - 3) - 3(x + 2) = 2x$

(g) $\frac{9y-12}{5} = 3$

(h) $4 - \frac{z-2}{2} = z + \frac{1-2z}{3}$

3. The result of doubling a number and adding 17 is the same as trebling it (multiplying by 3) and adding 4. What is the number?

4. A cup of tea costs 10p less than a coffee and a hot chocolate costs 20p more than a coffee. The cost of three coffees, five cups of tea and two cups of hot chocolate is £26.70.

- (a) Form an equation with x representing the price of a coffee.
- (b) Solve your equation to and hence determine the price of each beverage.

Extension

1. Solve the following equation:

$$8.9(x - 3.5) - 4.2(3x + 2.3) = 4.7x + 1.19$$

2. Solve the following equation:

$$\frac{1}{2}(1 - x) - \frac{1}{3}(2 + x) + \frac{1}{4}(3 - x) = 1$$